SELF-TEST

(Answers to Self-Test questions are in the Self-Test Appendix.)

1. Explain how the CPI is calculated.
2. What is a base year?
3. In year 1, your annual income is $45,000 and the CPI is 143.6; in year 2, your annual income is $51,232 and the CPI is 150.7. Has your real income risen, fallen, or remained constant? Explain your answer.

Measuring Unemployment

Every month, the government surveys thousands of households to gather information about labor market activities. It uses the information from the survey to derive the number of Americans unemployed.

Who Are the Unemployed?

The total population of the United States can be divided into two broad groups (Exhibit 5). One group consists of persons who are (1) under 16 years of age, (2) in the armed forces, or (3) institutionalized—that is, they are in a prison, mental institution, or home for the aged. The second group, which consists of all others in the total population, is called the *civilian noninstitutional population.*

The civilian noninstitutional population, in turn, can be divided into two groups: persons *not in the labor force* and persons in the *civilian labor force.* (Economists often refer to the "labor force" instead of the "civilian labor force.")

Civilian noninstitutional population = Persons not in the labor force + Persons in the labor force

Persons not in the labor force are neither working nor looking for work. For example, people who are retired, who are engaged in own-home housework, or who choose not to work fall into this category.

Persons in the civilian labor force fall into one of two categories: *employed* or *unemployed.*

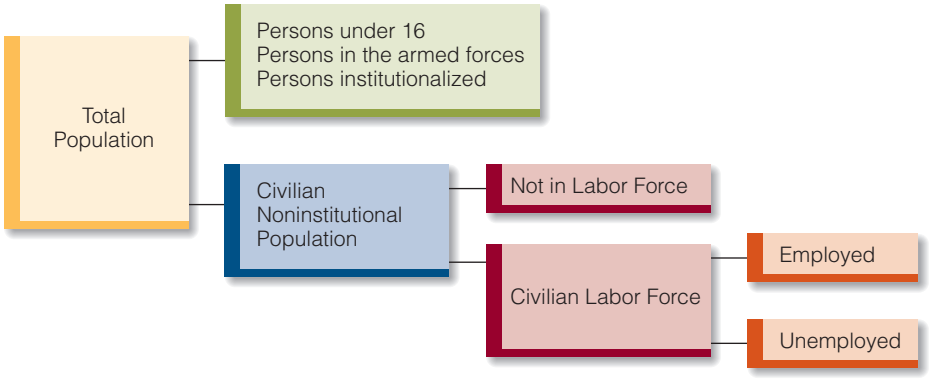
Civilian labor force = Employed persons + Unemployed persons

According to the BLS, employed persons consist of:

- All persons who did any work for pay or profit during the survey reference week.
- All persons who did at least 15 hours of unpaid work in a family-operated enterprise.

exhibit 5

Breakdown of the U.S. Population and the Labor Force



WHAT WAS A PENNY WORTH?

You are walking and you look down and see a penny. Do you stop and pick it up? Many people will say no, stating that it isn't worth their stopping and picking up a penny. After all, a penny is not worth very much, they say.

You buy a magazine at your campus store. The total price (including tax) comes to $6.02. You hand the cashier a $10 bill. Instead of giving you $3.98 in change, she reaches into the cup next to the cash register and pulls out two pennies; she gives you back $4 in change. It is easier for her to get the two pennies from the cup than to count out $3.98 in change. You say "thank you" and go on your way.

Another day comes, and this time you buy something for $5.99. You hand the cashier $6. When the cashier gives you back one penny in change, you drop the penny in the cup next to the cash register. One day you take from the cup, another day you give to the cup.

Twenty years ago, there was no penny cup next to cash registers. Why no penny cups then, but penny cups now? It's because 20 years ago a penny had more purchasing power than it does today. If a penny today had greater purchasing power than it does have, then you'd be less likely to drop a penny change into the penny cup; instead, you'd drop it in your pocket.

The discussion of cups and pennies gets us thinking. What was a penny worth, say, in 1900? A different way of asking the question: What coin today is equal to the purchasing power a penny had in 1900? The answer is a quarter. In other words, having a penny in 1900 was equivalent of having a quarter (25 cents today). It follows then that having four pennies in 1900 was the equivalent of having $1 today.

We leave you with the question we started out with, but this time we change the date. It is 1900 and you are walking along and you look down and see a penny. Do you stop and pick it up?

- All persons who were temporarily absent from their regular jobs because of illness, vacation, bad weather, industrial dispute, or various personal reasons.

According to the BLS, unemployed persons consist of:

- All persons who did not have jobs, made specific active efforts to find a job during the prior four weeks, and were available for work.
- All persons who were not working and were waiting to be called back to a job from which they had been temporarily laid off.

The Unemployment and Employment Rates

The **unemployment rate** is the percentage of the civilian labor force that is unemployed. It is equal to the number of unemployed persons divided by the civilian labor force.

$$\text{Unemployment rate } (U) = \frac{\text{Number of unemployed persons}}{\text{Civilian labor force}}$$

The **employment rate** (sometimes referred to as the *employment/population ratio*) is the percentage of the civilian noninstitutional population that is employed. It is equal to the number of employed persons divided by the civilian noninstitutional population:

$$\text{Employment rate } (E) = \frac{\text{Number of employed persons}}{\text{Civilian noninstitutional population}}$$

Finally, the **labor force participation rate** (LFPR) is the percentage of the civilian noninstitutional population that is in the civilian labor force.

$$\text{Labor force participation rate (LFPR)} = \frac{\text{Civilian labor force}}{\text{Civilian noninstitutional population}}$$

Unemployment Rate
The percentage of the civilian force that is unemployed: Unemployment rate = Number of unemployed persons/Civilian labor force.

Employment Rate
The percentage of the civilian noninstitutional population that is employed: Employment rate = Number of employed persons/Civilian noninstitutional population.

Labor Force Participation Rate
The percentage of the civilian noninstitutional population that is in the civilian labor force. Labor force participation rate = Civilian labor force/Civilian noninstitutional population.

The LFPR may sound like the employment rate, but it is different. Although the denominator in both is the same, the numerator in the employment rate is the number of employed persons, and the numerator in the LFPR is the civilian labor force (which consists of both employed persons and unemployed persons). For this reason, some economists say that while the employment rate gives us the percentage of the population that is working, the LFPR gives us the percentage of the population that is *willing to work.*

Common Misconceptions

About the Unemployment and Employment Rates

Many people mistakenly think that if the unemployment rate is, say, 7 percent, the employment rate must be 93 percent. They think this because they believe that the unemployment rate plus the employment rate must equal 100 percent. But the sum of the unemployment rate and the employment rate do not equal 100 percent. That's because the denominator of the unemployment rate is not the same as the denominator of the employment rate. The unemployment rate is a *percentage of the civilian labor force*. The employment rate is a *percentage of the civilian noninstitutional population,* which is a larger number than the civilian labor force.

Reasons for Unemployment

Usually, we think of an unemployed person as someone who has been fired or laid off from his or her job. Certainly, some unemployed persons fit this description, but not all of them do. According to the BLS, an unemployed person may fall into one of four categories.

1. *Job loser.* This is a person who was employed in the civilian labor force and was either fired or laid off. Most unemployed persons fall into this category.
2. *Job leaver.* This is a person employed in the civilian labor force who quits his or her job. For example, if Jim quit his job with company X and is looking for a better job, then he is a job leaver.
3. *Reentrant.* This is a person who was previously employed, hasn't worked for some time, and is currently reentering the labor force.
4. *New entrant.* This is a person who has never held a full-time job for two weeks or longer and is now in the civilian labor force looking for a job.

Unemployed persons = Job losers + Job leavers + Reentrants + New entrants

Discouraged Workers

Suppose Adam is fired from his job at company A in September. He looks for a job for about six months. During this time, he is considered an unemployed person and is counted in the unemployment rate. At the end of the sixth month, Adam is very discouraged; he doesn't think he will ever find a job, and so he stops looking. A month passes and he continues not to look for a job. Is Adam still considered an unemployed person? The answer is no. Remember, to be an unemployed person, you have to meet certain conditions, one of which is that you have to be actively looking for work. But Adam isn't actively looking for work, and he isn't waiting to be called back to a job or to report to a job. So Adam isn't unemployed. Because Adam is not unemployed, he does not get counted in the unemployment rate.

The BLS considers Adam a *discouraged worker.* You may think that for all practical purposes, a discouraged worker is the same as an unemployed person (because neither has a job). But they aren't the same for calculating the unemployment rate. The unemployed person gets counted, but the discouraged worker does not.

Some economists think that because discouraged workers are not considered unemployed, the unemployment rate is biased downward. Consequently, it doesn't really give us a good fix on the "real unemployment problem" in society.

Types of Unemployment

This section describes a few types of unemployment.

FRICTIONAL UNEMPLOYMENT Every day, demand conditions change in some markets, causing qualified individuals with transferable skills to leave some jobs and move to others. To illustrate, suppose there are two computer firms, A and B. For some reason, the demand falls for firm A's computers and the demand rises for firm B's computers. Consequently, firm A produces fewer computers. With fewer computers being produced, firm A doesn't need as many employees, so it fires some employees. On the other hand, firm B produces more computers. With more computers being produced, firm B hires additional employees. The employees fired from firm A have skills that they can transfer to firm B—after all, both firms produce computers. However, it takes time for people to transfer from one firm to another. During this time, they are said to be frictionally unemployed.

The unemployment owing to the natural "friction" of the economy, which is caused by changing market conditions and is represented by qualified individuals with transferable skills who change jobs, is called **frictional unemployment**. We use the symbol U_F to designate the frictional unemployment rate, which is the percentage of the labor force that is frictionally unemployed.

Frictional Unemployment
Unemployment due to the natural "friction" of the economy, which is caused by changing market conditions and is represented by qualified individuals with transferable skills who change jobs.

In a dynamic, changing economy like ours, there will always be frictional unemployment. Many economists believe that the basic cause of frictional unemployment is imperfect or incomplete information, which prevents individuals from leaving one job and finding another instantly.

Consider the situation where there are 1,000 job vacancies and 1,000 persons with the qualifications to fill the jobs. Will there be some unemployment? It is likely that there will be because not every one of the 1,000 job seekers will know where an available job is, nor will all employers give the job to the first applicant who knocks on the door (employers don't know if "better" applicants are around the corner). Matching qualified workers with jobs takes time.

STRUCTURAL UNEMPLOYMENT **Structural unemployment** is unemployment due to structural changes in the economy that eliminate some jobs and create other jobs for which the unemployed are unqualified. Most economists argue that structural unemployment is largely the consequence of automation (laborsaving devices) and long-lasting shifts in demand. The major difference between the frictionally unemployed and the structurally unemployed is that the latter do not have transferable skills. Their choice is between prolonged unemployment and retraining. For example, suppose there is a pool of unemployed automobile workers and a rising demand for computer analysts. If the automobile workers do not currently have the skills necessary to become computer analysts, they are structurally unemployed. We use the symbol U_S to designate the structural unemployment rate, which is the percentage of the labor force that is structurally unemployed.

Structural Unemployment
Unemployment due to structural changes in the economy that eliminate some jobs and create other jobs for which the unemployed are unqualified.

NATURAL UNEMPLOYMENT Adding the frictional unemployment rate and the structural unemployment rate gives the **natural unemployment** rate (or natural rate of unemployment). We use the symbol U_N to designate the natural unemployment rate. Currently, most economists estimate the natural unemployment rate at between 4 and 6.5 percent.

Natural Unemployment
Unemployment caused by frictional and structural factors in the economy. Natural unemployment rate = Frictional unemployment rate + Structural unemployment rate.

$$\text{Natural unemployment rate } (U_N) = \text{Frictional unemployment rate } (U_F) + \text{Structural unemployment rate } (U_S)$$

Cyclical Unemployment

The unemployment rate that exists in the economy is not always the natural rate. The difference between the existing unemployment rate and the natural unemployment rate is the **cyclical unemployment rate** (U_C).

Cyclical Unemployment Rate
The difference between the unemployment rate and the natural unemployment rate.

$$\text{Cyclical unemployment rate } (U_C) = \text{Unemployment rate } (U) - \text{Natural unemployment rate } (U_N)$$

When the unemployment rate (U) that exists in the economy is greater than the natural unemployment rate (U_N), the cyclical unemployment rate (U_C) is positive. For example, if $U = 8$ percent and $U_N = 5$ percent, then $U_C = 3$ percent. When the unemployment rate that exists in the economy is less than the natural unemployment rate, the cyclical unemployment rate is negative. For example, if $U = 4$ percent and $U_N = 5$ percent, then $U_C = -1$ percent.

Common Misconceptions

About Full Employment

What do you think of when you hear the term *full employment*? Most people think *full employment* means that the actual or reported unemployment rate is zero. But a dynamic, changing economy can never have full employment of this type due to the frictional and structural changes that continually occur. In fact, it is natural for some unemployment to exist—some natural unemployment, that is. For this reason, economists *do not* equate full employment with a zero unemployment rate. Instead, for economists, **full employment** *exists when the economy is operating at its natural unemployment rate.* For example, if the natural unemployment rate is 5 percent, then full employment exists when the unemployment rate (in the economy) is 5 percent. In other words, the economy can be operating at full employment, and some people will be unemployed.

Full Employment
The condition that exists when the unemployment rate is equal to the natural unemployment rate.

SELF-TEST

1. What is the major difference between a person who is frictionally unemployed and one who is structurally unemployed?
2. If the cyclical unemployment rate is positive, what does this imply?